Q05: Discussion Questions

DLS 2026 — Kukla

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1. Chain-of-thought helps on multi-step tasks but can lower accuracy on simple ones. What property of a task would you measure *in advance* to predict which regime you are in?

2. Best-of-n coverage is $1 - (1 - p_1)^n$, but realized accuracy depends on the verifier. Where do you invest a fixed budget: a better base model (raises p_1) or more samples (raises n)? What determines the answer?

3. If the written reasoning trace is not guaranteed to be faithful to the computation, is it ever safe to use the trace to decide whether to trust the answer? Argue a position.

4. "Overthinking" produces an inverted-U of accuracy vs. reasoning length. Design an inference-time controller that stops reasoning at the right point without knowing the answer.

5. Test-time compute can substitute for parameters on some benchmarks. For a product with a fixed latency budget, how would you decide between a bigger model and a smaller model that thinks longer?